

Caden Phillips

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MPhys Astrophysics student at the University of Liverpool, specialising in compact objects, time-domain astrophysics, and computational modelling. Research experience in Fisher matrix investigation, TOV solvers, hybrid equation of state construction, X-ray time-domain analysis of NICER data and spectral analysis of XMM-Newton data. Experienced in building and working with pipelines from numerical solutions to data processing, with strong Python and scientific computing skills.

EDUCATION

MPhys Astrophysics, University of Liverpool

Expected 2027

Projected First Class · Year 3: 73.5%

Academic Advisor: Dr. Christopher Copperwheat

Selected modules: Computational Modelling, Relativity and Cosmology, Advanced Observational Astrophysics, Physics of Galaxies, Physics of Planets

Full transcript available upon request.

RESEARCH EXPERIENCE

Internship — Institute of Nuclear Physics (IFJ PAN), Kraków, Poland *Jul 2025 - Aug 2025*

Supervisor: Dr. David Álvarez Castillo

Project: Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range

- Applied a Fisher Information Matrix framework to estimate tidal deformability uncertainties and assess error propagation in binary neutron star systems.
- Developed and implemented a Tolman-Oppenheimer-Volkoff (TOV) solver to compute mass-radius and tidal deformability relations for hybrid equations of state.
- Processed 62 hybrid equations of state, generating mass-radius and tidal deformability curves (approximately 400 stellar configurations per EoS) and analysing stable and unstable branches.
- Compared model predictions with GW170817 50% and 90% credible regions to assess equation-of-state viability.
- Co-presented results at the IFJ PAN Summer Internship Symposium 2025 with a fellow intern.

Field Work — Teide Observatory, Tenerife, Spain

Jun 2025

Supervisor: Dr. Daniel Harman

- Operated the 0.8 m IAC-80 telescope and a Celestron C14 Edge HD system with a QHY-163M/C CMOS detector.
- Identified and observed targets as a team, including M7, M13, Abell 39.
- Calibrated and reduced multi-band CCD data to construct Hertzsprung-Russell diagrams and composite images.

Independent Study — HEASARC/NICER Analysis Pipeline

Winter 2025 - Spring 2026

Self-directed observational analysis project

- Developed a Python pipeline integrating HEASoft tools via Bash to automatically download and organise archival NICER X-ray observations from the HEASARC database.
- Automated event filtering and calibration with scripted XSELECT workflows before light-curve extraction, across up to hundreds of ObsIDs.
- Produced calibrated X-ray light curves and diagnostic plots to enable time-domain comparison of source variability across observing epochs.

Independent Study — X-ray Weak Active Galactic Nuclei

Summer 2026

Self-directed project based on an internship offered by LEAPS Leiden ESA

- Built an end-to-end Python pipeline automating XMM-Newton EPIC data reduction (SAS), spectral fitting (PyXSPEC) and Bayesian hardness-ratio estimation (BEHR), applied to 90+ quasars at redshift 1.8 to 2.2 drawn from a crossmatch of the 5XMM-DR15 and SDSS DR16Q catalogues.
- Classified sources as X-ray weak, normal or strong via $\Delta\alpha_{\text{ox}}$, and examined how photon index, hardness ratio and fractional variability vary across the $\Delta\alpha_{\text{ox}}$ distribution, accounting for count-dependent fit reliability and sample selection bias.
- Validated the pipeline against published data.

RESEARCH OUTPUTS

- **Presentation:** *Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range*. IFJ PAN Summer Internship Symposium 2025. [Slides].
- **Report:** *Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range*. Institute of Nuclear Physics (IFJ PAN), Kraków, 2025. [PDF].
- **Pipeline:** *Using HEASoft to Analyse Archived NICER Data of tMSPs*. Independent Study, 2026. [Code].

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, Astropy), C++, Bash.
- **Tools & software:** RNS, LORENE, Jupyter, ROOT, Git, L^AT_EX, HEASoft.
- **Scientific methods:** TOV and hybrid EoS modelling, Fisher matrix estimation, numerical ODE solvers, gravitational wave parameter inference.

AWARDS & MEMBERSHIPS

- Awarded competitive summer internship placement at IFJ PAN (Poland) through the PHYS309 Internship Module. [PDF].
- Associate Member, Institute of Physics (IOP) *since 2023*.

INTERESTS

Astrophysics (compact objects, neutron stars, gravitational waves); astrophotography; hiking.